

Toy example: sensitivity of $L_\beta^{(k)}$ for small k . Place three points on the real line,

$$a = 0, \quad b = 1, \quad c = 0.25,$$

so $d^2(a, b) = 1$, $d^2(a, c) = \frac{1}{16}$, $d^2(b, c) = \frac{9}{16}$. With $k = 1$ the nearest-neighbor graph is $a \leftarrow c \rightarrow b$ (both endpoints claim c , and c claims a), giving local average squared radii

$$S_1(a) = \frac{1}{16}, \quad S_1(b) = \frac{9}{16}, \quad S_1(c) = \frac{1}{16},$$

and initial sampleability ratio $L_\beta^{(1)} = 9$.

Perturbation. Move c toward b by $\varepsilon > 0$, setting $c_\varepsilon = 0.25 + \varepsilon$. For $\varepsilon < 0.25$ the neighbor graph is unchanged, so

$$L_\beta^{(1)}(\varepsilon) = \frac{(0.75 - \varepsilon)^2}{(0.25 + \varepsilon)^2}.$$

Differentiating at $\varepsilon = 0$:

$$\left. \frac{d}{d\varepsilon} L_\beta^{(1)}(\varepsilon) \right|_{\varepsilon=0} = \frac{-2(0.75)[(0.25) + (0.75)]}{(0.25)^3} = -96.$$

A displacement of $\varepsilon = 0.01$ — one percent of the total arc length — changes $L_\beta^{(1)}$ by ≈ -0.96 , a **10% drop**.

Contrast with L_α . Since only pairs involving c are perturbed,

$$L_\alpha(X, Y_\varepsilon) = \frac{1}{9} \left[2|(0.25 + \varepsilon)^2 - (0.25)^2| + 2|(0.75 - \varepsilon)^2 - (0.75)^2| \right] = \frac{4\varepsilon}{9} + O(\varepsilon^2),$$

so $dL_\alpha/d\varepsilon|_{\varepsilon=0} = \frac{4}{9} \approx 0.44$. The sensitivity ratio is

$$\frac{|dL_\beta^{(1)}/d\varepsilon|}{|dL_\alpha/d\varepsilon|} = \frac{96}{4/9} = 216.$$

For the joint loss $J = L_\alpha + \lambda \log L_\beta^{(1)}$,

$$\left. \frac{dJ}{d\varepsilon} \right|_{\varepsilon=0} = \frac{4 - 96\lambda}{9},$$

which is dominated by the sampleability term even at modest λ . The step size η must therefore be calibrated to $O(1/96)$, not to L_α .

Discontinuity. At $\varepsilon = 0.25$ (i.e., c reaches the midpoint 0.5), the nearest neighbor of b switches from c to a , and $L_\beta^{(1)}$ drops to 1. This kink is the exact discontinuity that the smooth surrogate $\log \tilde{L}_{\beta, \tau}^{(k)}$ regularizes via the log-sum-exp approximation.

Why larger k cures this. With $k = 2$ on $N = 3$ points every $S_2(i)$ averages over both remaining neighbors, yielding $L_\beta^{(2)} = 1$ for all positions of c . More generally, averaging over k neighbors dilutes any single-displacement effect by $1/k$, and the Lipschitz constant in the metric stability theorem shrinks from $N^2/1$ to N^2/k .

ACKNOWLEDGMENT

Toy example to go along with the algorithm built for sampleability project (under mother project of synthetic data generation in generic settings, DTRA-inspired) .

REFERENCES

- [1] H. P. G. Lam, *Close-to-Geometry-Preserving Sampleable Generation from Labeled Point Clouds*, preprint, Worcester Polytechnic Institute, 2026.
- [2] H. P. G. Lam, *Sampleability Transport, Nonlinear Regularization, and the Porous Medium Flow*, preprint, Worcester Polytechnic Institute, 2025.

DEPARTMENT OF MATHEMATICAL SCIENCES, WORCESTER POLYTECHNIC INSTITUTE, WORCESTER, MA 01609
Email address: `h1am@wpi.edu`